

# Display lag effects on postural stability and cybersickness during active exposures to HMD virtual reality

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## ABSTRACT

This study examined whether a person's spontaneous postural sway before, and their head-movements during, exposure to virtual reality (VR) predicts their experiences of cybersickness. We compared the stability of head and body movements made by 50 HMD users to the sickness they experienced during VR simulations with different amounts of display lag. Consistent with *Postural Instability Theory*, we found that: 1) naturally unstable participants were significantly more likely to become sick during these laggy simulations; and 2) the severity of this sickness depended on the spatial magnitude and the temporal dynamics of their head movements during active HMD VR.

**Keywords:** Head-mounted display, Cybersickness, Motion-to-photon latency, Postural Instability.

**Index Terms:** Human-centered computing—Human computer interaction (HCI)—Virtual Reality—Empirical studies in HCI

## 1 INTRODUCTION

Whenever you move during HMD VR you risk becoming sick. Display lag appears to be one of the major triggers for this sickness [1]. But how does this lag cause cybersickness? And might it be possible to predict trial-to-trial and/or individual differences in the experience of this lag-induced sickness?

According to *Postural Instability Theory* (PIT) [2], motion sickness is caused by prolonged instability of either the body or its segments. It predicts that: 1) people with impaired postural control should be more susceptible to sickness; and 2) conditions that increase instability should result in more severe sickness. While there is some support for the first hypothesis [3, 4, 5], there has been little examination of the second hypothesis during HMD VR.

In order to test both hypotheses, we exposed 50 participants to HMD VR simulations with different amounts of artificially imposed display lag. We measured their spontaneous sway before entering VR, as well as their head-movements during seated VR, and their reports of cybersickness directly after leaving VR.

## 2 METHOD

### 2.1 Participants

31 females and 19 males were recruited from the University of Wollongong and the general population. They ranged in age from 18 to 45 years. All were without visual, neurological, or vestibular impairments, and had normal or corrected-to-normal vision.

## 2.2 Design and Materials

On each trial, participants made continuous nodding head movements while they viewed PCVR simulations with 0 (baseline), 63, 125, 188 or 250 ms of added display lag [The average effective baseline lag with this Oculus Rift S HMD was ~4 ms]. These head movements were timed to a metronome set at 0.5 Hz.

## 2.3 Procedure

Our participants' spontaneous postural stability was first assessed using a Bertec force plate. We recorded the spontaneous fluctuations in their centre of foot pressure (CoP) as they stood quietly in the lab with their eyes open. During each 60 s recording, anterior-posterior (y) and medial-lateral (x) CoP were sampled continuously at 500 Hz. Participants were then seated for their 5 VR exposure trials. On each 3-minute trial, they made continuous oscillatory pitch head-movements while viewing a 3-D virtual room simulation. If they became sick during the simulation, they were instructed to press the 'ctrl' key on the keyboard immediately (to terminate the simulation). Directly after each trial, we obtained: 1) a "yes/no" response to the question "do you feel sick?"; and 2) a *Fast Motion Sickness* (FMS) scale severity rating (from 0 = "Not sick at all" to 20 = "Frank sickness"). Participants were then given a 5-min rest before the next trial. This procedure was repeated until they had completed all trials.

## 2.4 Postural measures

Individual differences in spontaneous postural stability were assessed using pre-exposure CoP, whereas display lag effects on head stability were assessed using the tracking data from the HMD. In both cases, the spatial magnitude of the body/head sway was estimated to be the standard deviation of the time series data. Detrended fluctuation analyses (DFAs) were also conducted on these CoP and head-movement data to see how they evolved over time [6]. The scaling component ( $\alpha$ ) of these DFAs, provided non-redundant information about the self-similarity of these data over different time scales.

## 3 RESULTS

### 3.1 Spontaneous postural instability and sickness

The number of trials where each participant felt sick (0-5) in this study was modelled based on the standard deviation, and the DFA  $\alpha$ , of their pre-exposure CoP time series data. The following generalized linear model (Poisson) examined these relationships:

$$\text{NumSickTrials} \sim 1 + \text{stdCoPx} + \text{stdCoPy} + \alpha\text{CoPx} + \alpha\text{CoPy} + (1 | \text{Participant})$$

The model was found to account for 5.4% of the variability in the total number of sick trials reported by participants. Of the 4 CoP based measures, only stdCoPx contributed significantly to these

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predictions [ $z = -2.50, p = 0.012$ ]. Participants with less variability in their pre-exposure M/L sway were sick on more trials (Figure 1).

### 3.2 Predicting the probability of sickness based on head-movements made during VR

The probability of a participant feeling sick on a particular trial (0-1) was modelled based on the standard deviation, and the DFA  $\alpha$ , of their pitch head-orientation data during the last 30 s of that trial. This logistic generalized mixed model (which had random intercepts across participants) examined these relationships:

$$\text{Probability (Sick)} \sim 1 + \text{stdPitchHO} + \alpha\text{PitchHO} + (1 | \text{Participant})$$

The model accounted for 81% of the variability in sick/well reporting. We found a significant effect of stdPitchHO [ $z = 2.43, p = 0.015$ ], which indicated that the probability of a participant becoming sick on a trial increased when their pitch head-movements were more variable (Figure 2).

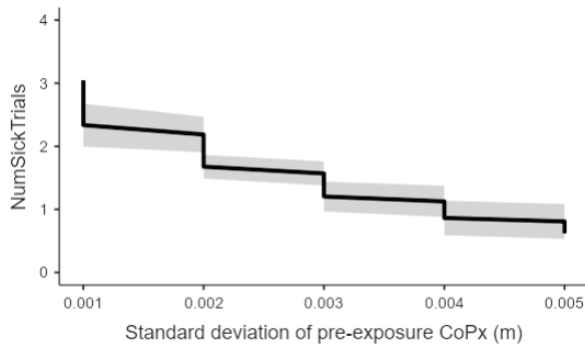


Figure 1: Relationship between stdCoPx and the total number of sick trials (shaded region is the standard error).

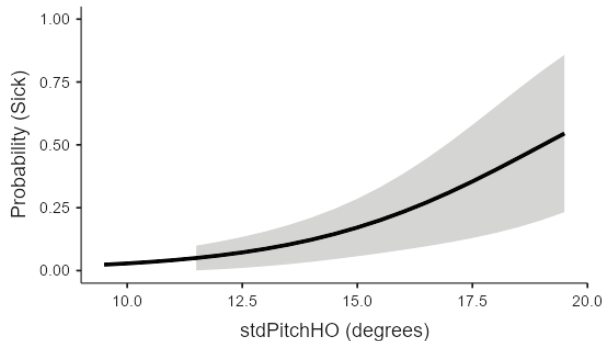


Figure 2: Relationship between stdPitchHO and the probability of becoming sick on a trial (shaded region is the standard error).

### 3.3 Predicting sickness severity based on head-movements made during VR

Sickness severity (0-20) during a VR trial was modelled in terms of the standard deviation, and the DFA  $\alpha$ , of the pitch head-orientation data for the last 30 s of that trial. The participant's Sick/Well status on the trial was also included in this linear mixed model (which had random intercepts across participants):

$$\text{Sickness Severity} \sim 1 + \text{stdPitchHO} + \alpha\text{PitchHO} + \text{sickStatus} + \text{sickStatus}:\text{stdPitchHO} + \text{sickStatus}:\alpha\text{PitchHO} + (1 | \text{Participant})$$

This model accounted for 76% of the variance in sickness severity ratings. We found a significant effect of stdPitchHO [ $F(1,239) =$

$7.540, p = 0.006$ ]. As can be seen in Figure 3, sickness severity increased with the standard deviation of pitch head-movements during VR. However, this effect of stdPitchHO on cybersickness was driven by the 'sick' trials [ $t(244) = 2.80, p = 0.006$ ]. It was not significant for the 'well' trials [ $t(239) = 1.02, p = 0.307$ ].

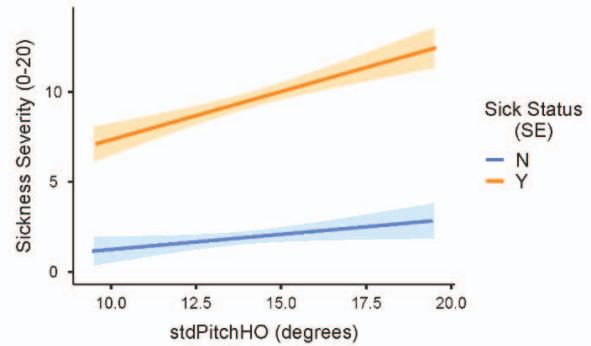


Figure 3: Relationships between stdPitchHO and sickness severity for 'Sick' (Y) and 'Well' (N) trials (shaded regions represent the standard error).

## 4 CONCLUSION

Consistent with the predictions of PIT, sick and well participants displayed different patterns of: 1) pre-exposure sway (standing in the real world), and 2) during-exposure head movements (seated in HMD VR). Surprisingly, participants who appeared more (not less) stable in the real world were more susceptible to cybersickness. However, consistent with PIT, VR conditions which produced more variable head movements were more likely to trigger sickness. These conditions were also found to generate more severe sickness (at least in susceptible participants). Taken together, our findings suggest that: 1) lag-induced sickness is caused and exacerbated by reductions in postural stability; and 2) software and hardware interventions for this sickness need to focus on promoting user stability in VR (as well as reducing its display lag).

## ACKNOWLEDGEMENTS

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